

BookBot Bin Logic Challenge

Retrieve by address, not by conveyor race

At a glance: Retrieve the right book by using bin addresses, not a conveyor race. Plan about 80 minutes for the core block; 4 to 6 teams of 4 to 5 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: automated storage and retrieval systems, bin addressing, routing rules, algorithmic search (Understand).
- Plan a legal orthogonal route that starts and finishes at DEPOT and retrieves every required address (Apply).
- Compare team legal moves with the verified minimum for the selected order card (Analyze).
- Defend a routing rule with the address order, legal move count, and card optimum (Evaluate).

MATERIALS FOR 25 STUDENTS AND 6 TEAMS

- bin cards
- address labels
- order deck
- planning route mats: one per team; shared judging mat: one
- timers
- optional grabber tool

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

- 01** Give each team a planning copy, then stage one shared tabletop judging mat. Mark DEPOT immediately left of A1 and run one team at a time.
- 02** Stage the order deck, staff answer key, score sheet, and verified minimum move count for every card.
- 03** Use tabletop pointers only. Keep the shared judging mat clear of actual library operations; no gloves, goggles, or trays are required.
- 04** Load the activity slides and ready the projected timer.

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Retrieve the right book by using bin addresses, not a conveyor race." Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Each team draws an order card and records a complete planned route from DEPOT back to DEPOT before its run.

Phase 3 Constraints

0:12 - 0:18

Set the route rules: start and finish at DEPOT; one move is an orthogonal step between adjacent bins or between DEPOT and A1; retrieve every requested address; no diagonal moves or jumps.

Phase 4 Run the shared-mat route

0:18 - 0:44

Teams plan on their copies, then execute the selected order card on the shared judging mat while staff count legal moves.

Phase 5 Test and record

Teams plan on their own copies, then run one at a time on the shared mat. Record legal-move count before time; a missed address or illegal jump makes the route incomplete. Use the instructor route and scoring key to compare the card minimum and normalize the efficiency score.

Phase 6 Algorithm defense

Teams compare their legal move count with the card minimum, compute the normalized efficiency score, and defend their routing rule.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, clear their pointer and order card, and reset the shared mat to DEPOT.

TROUBLESHOOTING

Problem: A route uses a diagonal or skips an address. **Fix:** Stop the run, mark it incomplete, and have the team trace a legal orthogonal route on its planning copy.

Problem: A move count is disputed. **Fix:** Replay the route slowly on the shared mat and count each adjacent square from DEPOT back to DEPOT.

Problem: Teams compare different cards directly. **Fix:** Normalize each result against that card's verified minimum before comparing efficiency.

Problem: A team finishes early. **Fix:** Give a second card for planning only and ask them to justify its route against the key.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to automated storage and retrieval.

Tier 2 (routing but not reasoning):

- "Which address order avoids unnecessary backtracking while keeping every move legal?"

Tier 3 (ready to extend):

- "Compare your legal move count with the card optimum and explain where any extra moves occurred."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Why store by address instead of by subject?
- What made one route use fewer legal moves?
- Where else is this storage logic used?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Correct retrievals	35
Legal complete route	25
Routing or algorithm explanation	20
Efficiency versus card optimum	10
Teamwork	10
Total	100

CLEANUP AND SOURCES

Return order cards, planning mats, labels, and timers; reset the shared judging mat and collect score slips.

SOURCES Temple Charles Library BookBot; Charles Library Circulating Collections